

flexhaz: Hazard-First Survival Distributions for R

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Summary

flexhaz is an R package for defining survival distributions through their hazard (failure rate) function. The user writes the hazard, any R function of time and parameters, and the package derives everything else. Given a hazard $h(t)$, the full distribution follows by numerical or analytical integration: cumulative hazard $H(t) = \int_0^t h(u) du$, survival $S(t) = \exp(-H(t))$, density $f(t) = h(t)S(t)$, quantiles, and random sampling. The package provides a complete likelihood model interface (log-likelihood, score, Hessian, and maximum likelihood estimation via `optim`) supporting exact, right-censored, and left-censored observations. Model diagnostics include Cox-Snell and Martingale residuals with Q-Q plots for goodness-of-fit assessment.

The central abstraction is the `dfr_dist` S3 class. All distribution methods return closures that accept time values and optional parameter overrides, enabling late binding of parameters for MLE:

```
library(flexhaz)
bathtub <- dfr_dist(
  rate = function(t, par, ...) {
    par[1] * exp(-par[2] * t) + par[3] + par[4] * t^par[5]
  },
  par = c(a = 0.5, b = 0.5, c = 0.01, d = 5e-5, k = 2.5)
)
S <- surv(bathtub)
S(10) # survival probability at t = 10

# Fit to censored data (delta: 1=exact, 0=right-censored)
df <- data.frame(t = rexp(50, 0.02), delta = sample(0:1, 50, TRUE))
solver <- fit(bathtub)
result <- solver(df, par = c(0.3, 0.3, 0.02, 1e-4, 2))
```

Statement of Need

Reliability engineers and biostatisticians who work with time-to-event data frequently encounter failure patterns that do not conform to standard parametric families. Bathtub curves, competing failure modes with additive hazards, and time-covariate interactions all require custom hazard specifications. Existing tools either restrict the user to a catalog of built-in distributions or require specifying the density and cumulative distribution function directly, quantities that are often harder to derive or reason about than the hazard itself. Researchers in reliability engineering and survival analysis need a framework that accepts a hazard function as the sole specification and provides the full distributional and inferential machinery automatically, including MLE with censored data and residual diagnostics.

State of the Field

Several R packages address parametric and semi-parametric survival modeling, but none accept a user-defined hazard function as the sole specification and derive the full distributional and inferential machinery.

Parametric frameworks. The survival package (Therneau, 2025) provides Kaplan-Meier estimation, Cox proportional hazards, and parametric accelerated failure time models, but does not support user-defined hazard functions. flexsurv (Jackson, 2016) allows user-defined distributions but requires the density or hazard together with its cumulative form, and focuses on regression modeling rather than standalone distribution objects with late-binding parameters. The eha package (Broström, 2024) supports proportional hazards and AFT models but is limited to built-in families.

Non-parametric and spline methods. muhaz (Hess & Gentleman, 2024) and bshazard (Rebora et al., 2014) estimate hazard functions non-parametrically via kernel smoothing and B-splines, respectively, but produce exploratory estimates rather than parametric distribution objects suitable for simulation or MLE. survPen (Fauvermier et al., 2019) fits penalized spline hazard models with automatic smoothing, and rstpm2 (Clements et al., 2024) implements Royston-Parmar models using restricted cubic splines on the log cumulative hazard. Both target data-driven flexible fitting rather than user-specified mechanistic hazard functions.

flexhaz fills the gap between these approaches. It accepts a single R function (the hazard) and derives the complete distribution, likelihood model, and diagnostics. The package integrates with the algebraic.dist (Towell, 2025a) distribution interface and the likelihood.model (Towell, 2025b) framework, yielding fisher_mle result objects with standard coef, vcov, confint, and summary methods.

Software Design

The dfr_dist constructor accepts a hazard rate function and optional analytical cumulative hazard, score, and Hessian functions. The resulting S3 object inherits from likelihood_model, univariate_dist, and dist classes. All distribution methods (hazard, surv, cdf, density, inv_cdf, sampler) return closures that accept time t , an optional parameter vector par , and additional arguments. This closure-returning pattern separates the distribution specification from parameter values, enabling the same object to serve as both a parameterized distribution for simulation and a template for MLE. Because additional arguments propagate through ... in every closure, the same architecture supports covariate-dependent hazards: $h(t, \text{par}, x, \dots)$ naturally yields covariate-conditional survival, density, and likelihood functions without special-case machinery.

The design follows a three-level optimization paradigm. At Level 1, the user supplies only the hazard function; cumulative hazard, score, and Hessian are all computed numerically. At Level 2, adding an analytical cumulative hazard eliminates numerical integration during likelihood evaluation. At Level 3, supplying analytical score and Hessian functions yields production-quality MLE with fast, accurate derivatives. Built-in constructors for exponential, Weibull, Gompertz, and log-logistic distributions provide Level 3 implementations. When analytical derivatives handle only exact and right-censored observations, the package automatically falls back to numDeriv::grad and numDeriv::hessian (Gilbert & Varadhan, 2019) for left-censored data, ensuring correctness without user intervention.

The fit method returns a solver closure: `solver <- fit(dist); result <- solver(df, par)`. The result is a fisher_mle object from the likelihood.model package, providing coef, vcov, confint, logLik, and summary methods.

Research Impact Statement

flexhaz is the foundation layer for an in-development family of packages addressing masked failure data in series systems. serieshaz composes dfr_dist components into series system distributions via additive hazards, and maskedhaz builds likelihood models for masked component-cause data with arbitrary hazard functions. A companion package maskedcauses, currently under CRAN review, provides closed-form solutions for exponential and Weibull components (798 tests, 99.45% coverage); maskedhaz results are validated against these analytical solutions to verify the numerical approach. The full ecosystem is available on r-universe at <https://queelius.r-universe.dev>.

AI Usage Disclosure

Claude Code (Anthropic) was used to assist with code refactoring, test generation, and drafting of documentation including this manuscript. All generated content was reviewed, edited, and validated by the author. The package design, mathematical formulations, and research decisions are entirely the author's own work.

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